Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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DRAFT - team details, declarations, team review and signature pending

1. TEAM AND PROJECT OVERVIEW

Team Name

Us!

Selected Domain

Domain 2: Orbyte Smart SME Challenge

Selected SDG Track (1–17)

8 - Decent Work and Economic Growth

Project / Solution Name

Orbyte DecisionOS

Name (Leader)

Awaiting team leader confirmation

Email & Phone (Leader)

Awaiting leader email and phone

University Affiliation(s)

Awaiting university affiliation(s)

Number of Team Members

Awaiting registered count (maximum 5)

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2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small cafes repeatedly check ingredient stock, request quotations and decide when to reorder. The owner often has to reconcile separate records before approving a purchase. Arithmetic mistakes, late ordering and undocumented decisions can disrupt service and add administrative work.

DecisionOS focuses on milk replenishment in a fictional cafe. The reference case has 18 litres in stock, estimated use of 12 litres per day and a three-day minimum coverage threshold. Current coverage is therefore 1.5 days. One supplier offers a lower price but estimates delivery after available stock runs out; another offers earlier delivery at a higher price. Price alone is insufficient for the decision.

The affected users are cafe owners and purchasing managers. The prototype brings stock evidence, delivery assumptions and quotation arithmetic into one transparent approval journey. It aims to reduce repetitive comparison work while preserving owner control and a traceable record.

This is a problem hypothesis supported by the challenge brief and a synthetic scenario, not a completed SME field study. A timed usability comparison is planned. No real customer, employee or company records are used, and no business savings or stockout reductions have been measured.

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2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Orbyte DecisionOS combines a working operational workflow with an optional AI scenario layer. A user initiates an inventory check; our mock Orbyte-style connector retrieves synthetic inventory and quotations. Python rules calculate coverage, cost and delivery risk. An owner selects or rejects an option, records a reason and receives a persistent purchase-order draft and audit trail after approval. No supplier is contacted and no payment is made.

The UI exposes the RM90 premium for earlier estimated milk delivery. Backend role checks, snapshot validation and idempotency protect approval and draft creation.

The optional MiroFish adapter explores a delivery delay or menu-price change through fictional stakeholder perspectives. Its source is pinned and HTTP contracts are tested. A separate environment is installed, and small Gemini API verification calls succeeded. A genuine run still needs a valid Zep key and a checked provider quota. The UI shows unavailable status rather than invented results. Agent narratives will not replace arithmetic or become unsupported forecasts.

The approach applies database, API and cloud-deployment concepts relevant to the workshops. All mandatory team workshops are complete. The project now runs in Replit; the approval workflow and reload persistence were verified there. The priority is a complete operational journey.

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3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16.4.2, React 19.3.1, TypeScript 5.9.3 and Tailwind CSS 4.3.3. Backend: Python 3.12, FastAPI 0.135.1, Pydantic 2.12.5 and SQLAlchemy 2.0.48. SQLite stores business records, decision snapshots, approvals, order drafts, audit events and simulation requests. Money uses integer sen; quantities and rules use Decimal.

Next.js proxies API requests to FastAPI. The mock connector is our own documented interface assumption, not an official Orbyte API. Optional simulation requests are processed by a separate local worker against MiroFish commit 7657031ac01184afe2cb220f5ee3545573b5e843. DecisionOS external execution remains disabled; the separate Gemini provider completed small verification calls, while Zep authentication remains incomplete.

In Replit, 33 backend tests, three component tests, TypeScript checks, the production build and four Playwright tests passed using installed Chromium 152. Native-browser checks verified approval and persistence. See Appendix A for architecture.

Replit: source imported and development preview verified on 23 October 2026. See the proof-of-use record. The team reports its Replit Core voucher expired before redemption; no Core credits are claimed. No GPU workload has been executed; RunPod is not applicable.

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3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The design starts with one recurring owner task: deciding which supplier can replenish a critical ingredient in time. A synthetic journey makes the data, trigger and decision explicit before adding broader modules. The interface places quote costs, estimated delivery and missing information beside owner approval controls; roles are clearly labelled as simulated.

The working local prototype has been iterated through automated backend/component checks and direct browser use. Approval, rejection, reload persistence and mobile navigation were checked. A timed study with consenting testers is still planned; no interviews or user validation are claimed.

Phase 2 work will validate the problem with SMEs, measure task completion against a manual baseline, and refine the workflow from observed errors. Real deployment would require authentication, tenant isolation, migrations, backups and a reviewed privacy process. The synthetic Replit development preview now runs and passed its approval and reload checks.

The main integration risk is unreliable or costly simulation execution. Deterministic procurement remains usable when MiroFish is offline. A small synthetic run will be measured before any AI result is demonstrated; the operational workflow remains usable without it.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8, targets 8.2 and 8.3. 8.2 concerns productivity through technological upgrading and innovation. DecisionOS aims to reduce repetitive procurement administration. 8.3 includes productive activities, entrepreneurship and SME growth. Traceable owner-controlled workflows are a proposed contribution, not evidence of policy-level impact. Source: UN DESA, Goal 8, https://sdgs.un.org/goals/goal8 (reviewed 22 Oct 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Estimated planning baseline: 10 minutes to complete a correct manual comparison, approval record and procurement draft for one shortage case. This is a team planning assumption, not an observed measurement or published benchmark. Replace it with the median of actual manual trials before making a measured improvement claim. Reference operational input: 18 / 12 = 1.5 days of stock coverage. This is calculated from synthetic data.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: percentage reduction in median time to complete a correct procurement draft, calculated as (manual median - DecisionOS median) / manual median x 100. Guardrails: arithmetic errors per task; percentage of approved drafts with actor, reason and timestamp; unassisted completion rate. Use identical synthetic tasks, alternate task order and report sample size and help given.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 target: at least 30% lower median task time; at a 10-minute baseline, this means no more than 7 minutes. Guardrails: zero reference-case arithmetic errors, 100% complete approval records and at least 80% unassisted pilot completion. These are proposed targets. No productivity gain, waste reduction, revenue change or stockout probability has been measured.

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5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [ ] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [ ] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [ ] All fields in this template have been completed within the word limits stated.

The user confirmed all team members completed the mandatory GenAI, Database and Cloud Deployment workshops. Other readiness boxes await registered-team review.

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6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No. The user reports that every registered member confirmed no pre-existing member-owned work was incorporated into this project. The DecisionOS application was created during this hackathon build after the official start. The separate MiroFish project and third-party frameworks remain their original owners' work and are disclosed in the Technical & Tools Declaration and Appendix B. Substantial Codex assistance is disclosed separately.

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6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js / React / Tailwind	Application frameworks	MIT
Python / SQLite	Language / database	PSF / public domain
FastAPI / Pydantic / SQLAlchemy	Backend frameworks	MIT
TypeScript / Playwright	Compiler / browser tests	Apache-2.0
MiroFish; other tools	Separate service; see Appendix B	AGPL-3.0; others as listed

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6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

No. The user reports that every registered member checked and confirmed no employment, internship or consulting agreement affects ownership of this hackathon work. The team leader should verify this answer again before signing the official declaration.

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7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]

Awaiting registered-team review and leader confirmation. No legal declaration or signature has been assumed.

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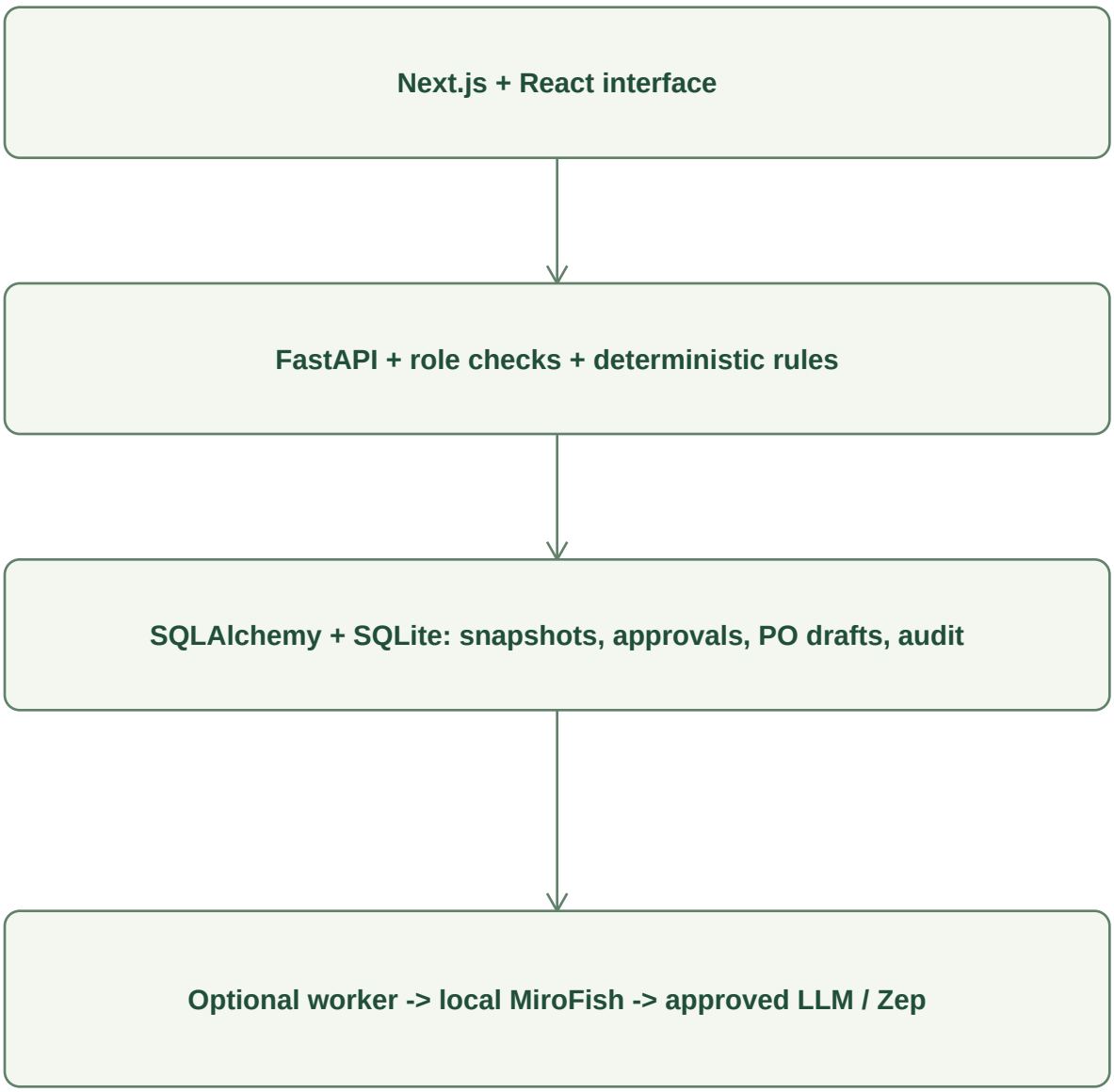
Appendix A: system architecture

Synthetic operational workflow; optional simulation remains disabled.



Our mock Orbyte-style connector supplies business-scoped synthetic inventory and quotes. No official Orbyte API is claimed.

Approval, purchase-order draft and audit records commit together. No real order, payment or inventory receipt occurs.

Replit development preview verified on 23 October 2026. The approval workflow, reload persistence and four Playwright tests passed. See the separate proof-of-use record.

Appendix B: dependency notices

Python 3.12 / SQLite	PSF / public domain
Next.js 16.4.2 / React 19.3.1 / React DOM 19.3.1	MIT
TypeScript 5.9.3 / Playwright 1.64.1	Apache-2.0
Tailwind CSS / PostCSS plugin 4.3.3	MIT
Lucide React 0.468.0	ISC
FastAPI 0.135.1 / Pydantic 2.12.5 / SQLAlchemy 2.0.48	MIT
Uvicorn 0.41.0 / HTTPX 0.28.1 / python-dotenv 1.2.2	BSD-3-Clause
Pytest 9.0.2 / Vitest 5.0.5 / jsdom 30.1.3	MIT
Testing Library React 16.3.3 / jest-dom 7.0.1	MIT
pytest-asyncio 1.3.0	Apache-2.0
MiroFish 7657031 (separate optional service)	AGPL-3.0

Versions and licences above were checked against installed package metadata and the pinned upstream licence. Full transitive dependency versions are retained in the lockfiles. MiroFish source is not bundled with the Apache-2.0 application. Preserve upstream notices and review AGPL obligations before external deployment.

Development assistance: OpenAI Codex (major use). Whisper base.en via faster-whisper transcribed workshop videos locally for rules review. ReportLab/PyPDF and the Codex presentation runtime generated documents. These research/authoring tools are not application services.